

Functional Requirements Document (FRD)

Retail Shelf Monitoring Using Computer Vision and Data Science

1. Purpose

This document defines the functional requirements for the Retail Shelf Monitoring system — an application that uses computer vision to analyze shelf images, detect and count products, identify empty/low-stock areas, flag misplaced items, and present results via a dashboard.

2. Scope

The system covers: image input, product detection, product counting, empty-shelf detection, misplacement detection, alert generation, and dashboard visualization/reporting. It does not cover real-time CCTV integration, POS/inventory system integration, or multi-store analytics (these are future enhancements).

3. Actors

Actor	Description
Store Employee/User	Uploads shelf image, views results, exports reports
System/Admin (optional)	Manages product categories, retrains/updates model
System (automated)	Runs detection pipeline, generates alerts

4. Functional Requirements

FR-1: Image Input

- FR-1.1 — System shall allow the user to upload a shelf image (JPG/PNG) via the dashboard.
- FR-1.2 — System shall validate the uploaded file format and size before processing.
- FR-1.3 — System shall display the uploaded image on the dashboard for reference.

FR-2: Image Preprocessing

- FR-2.1 — System shall resize the uploaded image to the model's required input dimensions.
- FR-2.2 — System shall normalize pixel values before passing the image to the detection model.

FR-3: Product Detection

- FR-3.1 — System shall run the YOLOv8n object detection model on the preprocessed image.
- FR-3.2 — System shall return bounding boxes, class labels, and confidence scores for each detected product.
- FR-3.3 — System shall discard detections below a configurable confidence threshold.
- FR-3.4 — System shall overlay bounding boxes and labels on the displayed image.

FR-4: Product Counting

- FR-4.1 — System shall count the number of detected products per category/class.
- FR-4.2 — System shall display a summary table of product counts by category.

FR-5: Empty/Low-Stock Detection

- FR-5.1 — System shall analyze shelf regions without detected products to identify empty spaces.
- FR-5.2 — System shall calculate a stock level or empty-space percentage per shelf zone.
- FR-5.3 — System shall flag any zone below a configurable stock threshold as low stock or empty.

FR-6: Misplaced Product Detection

- FR-6.1 — System shall compare detected product categories against the expected category for each shelf zone.
- FR-6.2 — System shall flag products detected in an incorrect zone as misplaced.
- FR-6.3 — System shall highlight misplaced items visually, for example with a different bounding-box color.

FR-7: Alerts & Shelf Status

- FR-7.1 — System shall generate a list of alerts (empty zones, low stock, misplaced items) after analysis.
- FR-7.2 — System shall compute an overall shelf-health score/status such as Good, Needs Attention, or Critical.

FR-8: Dashboard & Visualization

- FR-8.1 — System shall display the annotated image, product counts, and alerts in a single dashboard view.
- FR-8.2 — System shall provide charts (bar/pie) summarizing product distribution and stock status.
- FR-8.3 — Dashboard shall update results dynamically when a new image is uploaded.

FR-9: Reporting & Export

- FR-9.1 — System shall allow the user to export detection results as a CSV file.
- FR-9.2 — System shall allow the user to download the annotated image.

FR-10: Optional Natural-Language Summary

- FR-10.1 — System may generate a text summary of shelf status using a local LLM, e.g., "Zone B is 40% empty; 3 items misplaced."

5. Non-Functional Requirements

Category	Requirement
Performance	Detection + analysis should complete within a few seconds per image on a standard laptop CPU/GPU.
Usability	Dashboard should be simple and require no technical training to operate.
Portability	System should run locally without requiring paid cloud services.
Reliability	System should handle invalid/corrupt image uploads gracefully with an error message.
Scalability (future)	Architecture should allow extension to batch/multi-image or real-time video processing.

6. Assumptions & Constraints

- Model accuracy depends on the quality and size of the training dataset used.
- System is designed for single-image analysis in the MVP; not real-time video by default.
- All tools/models used are free and open-source, runnable on a standard laptop.
- Product category zones for misplacement detection must be predefined/configured.

7. Dependencies

- Pretrained/fine-tuned YOLOv8n model weights
- Annotated dataset for training and validation
- Python environment with OpenCV, Ultralytics, Streamlit, and Pandas installed